

MA.7.AR.4.2**Benchmark**

MA.7.AR.4.2 Determine the constant of proportionality within a mathematical or real-world context given a table, graph or written description of a proportional relationship.

Example: A graph has a line that goes through the origin and the point (5, 2). This represents a proportional relationship and the constant of proportionality is $\frac{2}{5}$.

Example: Gina works as a babysitter and earns \$9 per hour. She can only work 6 hours this week. Gina wants to know how much money she will make. Gina can use the equation $e = 9h$, where e is the amount of money earned, h is the number of hours worked and 9 is the constant of proportionality.

Connecting Benchmarks/Horizontal Alignment

- MA.7.AR.3.3
- MA.7.GR.1.3, MA.7.GR.1.5
- MA.7.GR.3.3
- MA.8.AR.3.2

Terms from the K-12 Glossary

- Constant of Proportionality
- Origin
- Proportional Relationships

Vertical Alignment**Previous Benchmarks**

- MA.6.AR.3.2
- MA.7.NSO.2.1, MA.7.NSO.2.2, MA.7.NSO.2.3
- MA.7.AR.3.2

Next Benchmarks

- MA.912.AR.2.2

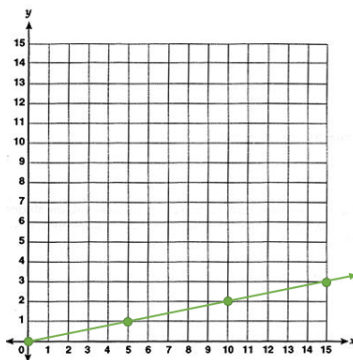
Purpose and Instructional Strategies

In previous courses, students determined rates and unit rates in ratios. In grade 7 accelerated, students take a broader view of a rate or unit rate as they understand it to be the constant of proportionality in a proportional relationship. Students will also expand their understanding of the constant of proportionality in proportional relationships to slope in linear relationships. In Algebra1, students write linear two-variable equations in all forms from real-world and mathematical contexts.

- Instruction includes different ways of representing proportional relationships, such as tables and graphs. Multiplying or dividing one quantity in a ratio by a particular factor requires doing the same with the other quantity in the ratio to maintain the proportional relationship. Graphing equivalent ratios create a straight line passing through the origin.
 - Tables

<i>x</i>	5	40	65	
<i>y</i>	1	8	13	

○ Graphs



- Starting instruction with real-world context rather than mathematical procedure allows students to reason through the meaning of the constant of proportionality (*MTR.7.1*). Connect prior knowledge of unit rates when developing the constant of proportionality.
- Instruction includes a connection to pi (π) as the constant of proportionality in the circumference formula within MA.7.GR.1.3.
- Problem types include positive and negative constants of proportionality.

Common Misconceptions or Errors

- Some students reverse the order of the ratio between the two quantities in a proportional relationship.
- Students may neglect the scale(s) of the axes on a graph. To address this misconception, have students interpret the constant of proportionality in context and evaluate the reasonableness of the answer. This may prompt students to revisit the graphical representation for better details.

Strategies to Support Tiered Instruction

- Instruction focuses on students' comprehension of the context or situation by engaging in questions (laminating these questions on a printed card for students to utilize as a resource in and out of the classroom would be helpful).
 - What do you know from the problem?
 - What is the problem asking you to find?
 - What are the two quantities in this problem?
 - How are the quantities related to each other?
- When determining the constant of proportionality in a graph, the teacher can instruct students to interpret the coordinate point as it relates to the titles of each axis. Teacher and students can use this information to co-create a table to clear up the misconception of misinterpreting the context of the constant proportionality.

- Instruction includes the use of a three-read strategy. Students read the problem three different times, each with a different purpose (laminating these questions on a printed card for students to utilize as a resource in and out of the classroom would be helpful).
 - First, read the problem with the purpose of answering the question: What is the problem, context, or story about?
 - Second, read the problem with the purpose of answering the question: What are we trying to find out?
 - Third, read the problem with the purpose of answering the question: What information is important in the problem?
- Teacher has students interpret the constant of proportionality in context and evaluate the reasonableness of the answer. This may prompt students to revisit the graphical representation for better details.
- Teacher revisits the development of the equation $y = px$ and be sure students see all proportional relationships have the origin as a common point.

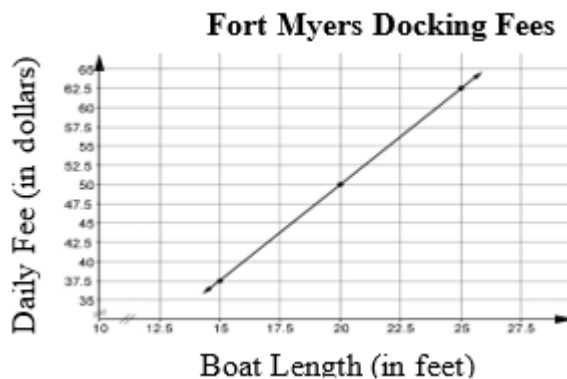
Instructional Tasks

Instructional Task 1 (MTR.5.1, MTR.7.1)

- Part A. The daily fee for docking a boat at a marina in Tarpon is proportional to the length of the boat. The table displays the fee for four different boat lengths. Find the constant of proportionality and explain what it means in the context of this problem.

Port Tarpon Docking Fees	
Boat Length (in feet)	Daily Fee (in dollars)
15 feet	\$33.75
17 feet	\$38.25
20 feet	\$45.00
21 feet	\$47.25

- Part B. The daily fee for docking a boat at a marina in Fort Myers is also proportional to the length of the boat. The graph displays the relationship between the fee and the boat length. Find the constant of proportionality and explain what it means in the context of this problem.



- Part C. At which marina is it less expensive to dock a boat? Explain how you determined your answer.

Instructional Items*Instructional Item 1*

After a workout at the gym, three friends made protein shakes to help in their recovery. Each protein shake contains 2 scoops of protein powder and 12 ounces of water. What is the constant of proportionality for this relationship?

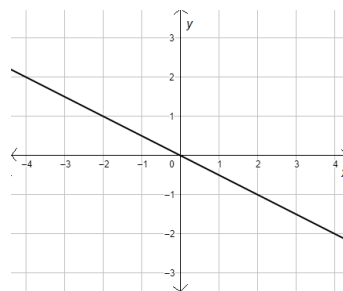
Instructional Item 2

Determine the constant of proportionality for the following proportional relationships.

1.

x	3	4	6	9
y	4.5	6	9	13.5

2.



**The strategies, tasks and items included in the BIG-M are examples and should not be considered comprehensive.*